

■# FLCR-32 - QCD And Color Transport In LCR

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper formalizes QCD/color language as a translation of finite face transport. The operative object is QCD color translation. The core result is that finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD. The paper also defines how this result routes forward: FLCR-32 cites FLCR-14 before making any color-transport claim. Its main residue is explicit: physical quark confinement, CKM, and measured QCD parameters remain external calibration.

Keywords

QCD color translation; LCR; receipt; claim lane; normal form.

External Reader Orientation

An outside reviewer should read this paper as a translation and comparator paper. Its local object is **QCD And Color Transport In LCR**. The paper's immediate contribution is: Maps QCD/color transport after algebraic face transport is established.

The nearest external anchors are standard mathematical physics definitions, cited Standard Model targets, calibrated constants or data where available, and the LCR-native papers that provide the source-side object. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It may close the external target definition and the internal translation grammar, but it does not claim measured physical identity unless a calibration/comparator row is present.

Reviewer Compression

Core object. QCD And Color Transport In LCR.

Primary result. This paper contributes a controlled mapping between a cited external target and a cited LCR-native source object.

Primary non-result. This paper does not claim measured physical identity unless the comparator/calibration row is attached.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, external_target_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript correctly treats Standard Model language as an external target surface and keeps translation claims separate from measured physical identity.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add a comparator table mapping each external target object to the exact LCR source object, invariant, calibration status, and falsifier.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.
- State scale, scheme, and units for any physics-facing numerical comparator.

Acceptance Blockers

- Measured physical identity cannot be accepted without comparator/calibration rows.
- The mapping from external target to LCR source must be explicit enough that another reviewer could reject or reproduce it.

1. Contribution And Scope

- Defines QCD color translation as a first-class FLCR object.
- States the local result: finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD.
- Routes downstream use through claim lanes rather than inherited prose: FLCR-32 cites FLCR-14 before making any color-transport claim.
- Preserves the residue boundary: physical quark confinement, CKM, and measured QCD parameters remain external calibration.

This paper belongs to the Standard Model translation band. Standard Model language names an external target surface, not an automatic LCR identity. A translation claim is admissible only when the cited standard object, the LCR-native source object, the mapping rule, the evidence lane, and the falsifier or calibration boundary are all visible.

2. Source Routing

Primary routed sources:

- `13_Standard_Model_Quark_Face_Transport.md`
- `supplements/SM_BRIDGE_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Structural color.** The finite face label before measured QCD identity.
- **QCD calibration.** The external data/theorem lane needed for physical QCD claims.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

4. Formal Claims

Claim	Lane	Statement
Theorem 32.1	standard_theorem_citation_bound_result	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD
Proposition 32.2	normal_form_result	QCD color translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 32.3	falsifier_or_open_obligation	physical quark confinement, CKM, and measured QCD parameters remain external calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 32.1	external target or comparator	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Proposition 32.2	translation grammar	QCD color translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 32.3	obligation/falsifier	physical quark confinement, CKM, and measured QCD parameters remain external calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR - 32 defines or uses QCD And Color Transport In LCR as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-32.

- 2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.
- 3. State the strongest admissible claim in its current lane.
- 4. Export only the lane-safe result to downstream papers and preserve the residue.

Proof Sketch

The proof strategy for FLCR-32 is a typed construction argument. The paper names structural color as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is QCD And Color Transport In LCR. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Legacy paper body	Primary formal and source-integration material assigned by the series map.
Internal and pyramid supplements	Cross-paper reasoning windows, deferred back-application slots, and route constraints.
NSIT tool closure matrix	Existing verifier/tool families that can close internal or batch claims.
CAM/crystal and standards-window evidence	Projected memory, source routing, standards reports, and window/node databases.

Standard Model Definition Dependencies

This paper consumes the dedicated Standard Model Defining layer. These papers define the external Standard Model or adjacent standard-physics object before this FLCR paper translates it into LCR language.

SMD paper	Definition surface	Required use
SMD -01	Standard Model Definition Contract And Evidence Boundary	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -03	Gauge Group SU3 SU2 U1 And Representation Content	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -05	QCD Color Dynamics And Hadronic Boundary	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -08	Renormalization, Effective Field Theory, And Running Parameters	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -12	Standard Model Comparator And Falsifier Protocol	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.

This paper also imports SMB-01, the Standard Model Closure Bridge. SMB-01 supplies the closure-by-lane rule: rows are no longer treated as unresolved by default; they are closed when standard formalism, FLCR proof, CAM/crystal reapplication, normal-form translation, or external calibration satisfies the lane, and they remain obligations only when a required field is missing.

Closed Rows Applied In This Paper

Row	Closure	Evidence	Boundary
QCD as $SU(3)_C$ non-Abelian gauge sector	closed by standard formalism	SMD-05, PDG source registry	Closes QCD definition, not LCR derivation of QCD phenomenology.
Structural color-face transport	closed by FLCR structural proof scope	FLCR-14, Paper 13 quark-face route, SMD-03	Closes finite face/color transport as structural algebraic correspondence.
Confinement/hadron spectra	not closed here	SMD-05, SMD-08	Requires QCD-specific evidence, lattice/phenomenological data, or calibrated comparator.

The paper can now state a stronger, cleaner result: FLCR already has a finite structural color-transport surface, and QCD already has a citable external $SU(3)_C$ definition. What remains open is the calibrated physical promotion from structural face transport to measured QCD dynamics.

Active Comparator Rows

Comparator	External side	LCR side	Current result	Next required action
Color representation	$SU(3)_C$ color representation	finite face/color transport	closed structurally	Bind <code>verify_quark_face_transport_literalized.py</code> receipt in final proof package.
Non-Abelian dynamics	field strength/self-interaction	no full LCR dynamics row yet	open/calibration-bound	Define dynamical equation or keep as external formalism only.
Confinement	QCD nonperturbative phenomenon	none proven	open	Requires explicit QCD comparator or dataset.

Executable Evidence Bound In This Pass

This paper now imports the quark-face receipts as the finite structural core of the QCD translation row. The important move is narrow and powerful: the receipt closes a color-transport-like algebraic surface, while the paper keeps confinement dynamics, running coupling, hadron spectra, and measured QCD phenomenology outside the closed row.

Evidence	Path	Result imported here	Boundary preserved
Quark-face literalization receipt	local evidence artifact	10/10 PASS; exact structural color transport through QuarkFaceForge, including S3 Weyl behavior, trace conservation, chirality doublet plus singlet, and neutral-extreme confinement check.	A structural color transport proof is not a full QCD derivation.
Quark-face transport receipt	local evidence artifact	Shell-2 has three chart states; trace-2 J3(O) idempotents close; S3 has six permutations and closes the trace-2 triple; n=3 SU(3) group-ring residual is exactly zero.	The receipt itself names physical quark color, CKM phase, V-A weak structure, and cold-start depth-only derivation as open layers.
External QCD definition surface	SMD-05, SMD-08, PDG registry	Supplies QCD/color dynamics and running-parameter targets for comparison.	Standard formalism defines the target; the FLCR receipt supplies the internal structural comparator.

The upgraded claim is: the color-transport comparator row is closed at the finite algebraic/receipt level. The remaining QCD work is no longer a vague open problem; it is the bounded task of comparing the closed finite transport surface against the external QCD objects declared by SMD-05 and SMD-08.

Online Source Rows Applied In This Pass

Online source	Claim-lane effect	Boundary retained
IRL -PDG-2026	Supplies the current external QCD/color-sector review target and evaluated particle-data context.	Does not convert finite quark-face transport into full QCD dynamics.
IRL -PDG-2024	Supplies a stable citable PDG review DOI for publication references.	Hadron spectra, confinement dynamics, and running coupling remain external comparator/calibration rows.

Online data therefore closes the external QCD target-definition problem. The internal FLCR row remains the quark-face receipt; the physical-dynamics row remains comparator-bound.

Assertive Closure Promotions

Row	Closure now allowed	Evidence	Residue moved out of the open row
QCD/color external target	closed by citation/definition	IRL - PDG - 2026, SMD - 05, SMD - 08	LCR identity with full QCD dynamics.
Structural color transport	closed internally at finite layer	quark_face_transport_receipt.json, quark_face_transport_literalized_receipt.json	Measured quark identity, confinement dynamics, hadron spectra.
SU(3) color-count comparator	closed as structural match	quark-face receipt plus PDG/QCD target	Running coupling and physical spectra remain calibration-bound.

The non-timid conclusion is: color-transport closure is already solved at the structural FLCR layer. What remains is not “do we have color transport,” but “what physical QCD rows can be calibrated against it.”

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GLM-OBLIGATION-CLOSURE	The GLM closure matrix independently compares verifier findings to named obligations and advances 11 obligations to partial closure while identifying the untouched set.	When fed back through the claim lanes, the matrix stops the suite from carrying stale open labels where bounded verifier evidence already exists, while preserving untouched obligations as active next channels.	Obligation status promotion becomes auditable instead of rhetorical.; Papers 09, 10, 13, 16, 18, and 33 gain bounded closure support from specific verifier findings.; The suite gains a clean split between partial closure, engineering scope, and untouched physical/calibration bridges.	Partial closure is not unconditional closure.; CKM, quark/color measurement, Higgs/QFT mass calibration, GR curvature bridges, and full Moonshine remain separate dependency lanes where the matrix says they remain open.

Combined Recursive Effects

Combination	Result	Solves
COMBO-APPLIED-PHYSICS-BOUNDARY	Applied forge topology supplies the domain boundary, GLM closure supplies bounded mathematical promotions, and address horizons supply finite computational evidence.	This lets later Standard Model and physics-facing papers state strong bounded claims while separating calibration-dependent physical identity claims.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GLM-OBLIGATION-CLOSURE	normal_form_result	Adds a closure matrix that compares independent GLM verifier findings against open obligations and records bounded promotions instead of leaving the papers in their earlier unresolved posture. Evidence facts: 12 findings considered against 29 obligations; 11 obligations advanced to partial; 17 untouched; closed-bounded findings include GLM-F1-SM01, GLM-F2-SP002, GLM-F4-RULE30P3, GLM-F6-YM-EXT02, GLM-F7-SM04, GLM-F8-HODGE-EXT01.	Promotions are lane-bound and partial where the matrix says partial; external physical bridges remain separate calibration obligations.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-R30-LATTICE-THEOREM-REGISTRY	standard_theorem_citation_bound_result	Adds executable theorem registry rows for octonions, J3(O), chart-to-J3(O), exact n=3 SU(3) Weyl closure, Rule-30 8x8 transition, Niemeier routes, relational qubit closure, Monster/Pariah boundary, and computational ISA signatures. Verification: Targeted pytest passed: 11 Rule-30 normal-form/proof-shell tests, including address decomposition, centroid state, proof-shell dependency acyclicity, Niemeier obligation boundaries, scoped claim promotion, and full dependency closure.	Imports verifier-backed internal/formal results; transported physical interpretations remain lane-bound.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.508566.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
carrier enumeration	normal_form_result	The LCR carrier action is closed as an addressable finite-state surface for its declared domain.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-4 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.509077 and mass sum=40.726174.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-02 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-02, FLCR-12, FLCR-22, FLCR-32; avg TarPit mass=0.511700; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
QCD color transport	receipt_bound_internal_result	QCD-facing color transport is closed structurally through quark-face receipts.	Full QCD dynamics remain calibration/comparator-bound.

Claim Posture After This Pass

FLCR- 32 should now be read as a resolved-state contribution for QCD And Color Transport In LCR at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR- 11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.

- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.

- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- physical quark confinement, CKM, and measured QCD parameters remain external calibration
- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 13/fold_action.

Original slot	Family	Lift	Role	Proof form
13	fold_action	1	order-2 slot-3: fold the initial action into a higher-dimensional representation	fold map, coordinate atlas, and representation transport

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport).

It must produce: formal result of original slot 13 plus the same-family lift path toward slot 23.

Semantic transition: fold the local state into a higher representation while preserving addressability.

Accepted formalism targets to bind in the journal rewrite:

- representation theory
- group actions and equivariance
- tensor or coordinate atlases
- transport maps between equivalent descriptions

Slot	Family	Lift	Produced proof form
13	fold_action	1	fold map, coordinate atlas, and representation transport

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to strengthen representation transport: every face, gauge, atlas, or fold label must point to a preserved source object.

Primary Evidence Inputs

Source	Placement reason
13_Standard_Model_Quark_Face_Transport.md	primary assigned source for the paper's formal spine
supplements/13_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations
virtuous\13_Standard-Model_Quark-Face_Transport_VIRTUOUS.md	prior enriched paper body; should be mined for mature claims and proof ordering

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/SM_BRIDGE_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.

- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

13	92	internally closed, external calibration pending	CL affirmative verifier: exact SU(3) color transport, QuarkFaceForge 10/10, Paper 03 trace/face support	CKM and measured quark/color bridge dataset
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Paper-Specific NSIT Closure Rows

Paper 13 Standard-Model quark-face transport	batch_tool_unison_closure	SU(3) T4 closure, D12 trace conservation, side-flip chirality, J3(O) faces, QuarkFaceForge	Paper 13	quark_face_transport_literalized_receipt.json 10/10	Exact structural color-transport claim closes; physical-quark identity remains transport-scoped.
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Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Symbolic Carrier Versus Physical Carrier	external_calibration_result	Promote addressable symbolic-carrier and transport claims where the paper names carrier, path, residue, or traversal rules.	Map every physical carrier phrase to a standard physics object, Hamiltonian/model, dataset, or calibration route before treating it as measured physics.	Physical identity, units, material constants, and measured transport remain calibration-bound.
CKM/Standard-Model Structural Transport Split	standard_theorem_citation_bound_result	Promote SU(3)/D12/J3 face transport as structural algebraic transport when the finite receipt is attached.	Attach quark-face transport receipt and keep CKM/PDG numerical values as data-binding requirements.	Measured CKM agreement, physical-quark identity, and parameter calibration remain data-bound.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Symbolic Vs Physical Carrier	This paper may use carrier language for addressable symbolic transport inside the LCR/CAM system.	Standard physics carrier terminology, local verifier receipts, material/plasma/transport supplements, and calibration routes.	Split symbolic carrier closure from measured physical carrier identity.	Physical identity, units, Hamiltonians, datasets, and measured constants remain external-calibration claims.
Ckm Sm Transport Split	This paper may claim structural face/color transport at the algebraic representation level.	SU(3)/D12/J3 receipts, quark-face transport verifier, and Standard Model/CKM data routes.	Promote structural transport while keeping numerical CKM/PDG agreement in data-binding lanes.	Physical-quark identity, measured parameters, and calibration remain citation/data-bound.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-13.50: Paper 13.50 - Quark-Face Transport Claim Contract	101	Paper 13.50 exists to keep the most tempting overclaim visible. The proved object is strong because it is exact and finite. The physical identification becomes stronger only when it is separately derived, not when it ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 11 — C-Form: Theory Admission Gate Gluon	77	C = the admission filter Gluon — the trust anchor that filters external theories by Gluon mass matching. In the lattice_forge substrate, C is realized as the admission gate that: - Takes an external theory's G...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-13.25: Paper 13.25 - Toolkit for Quark-Face Transport	77	This support paper exposes the tools for Paper 13. It is not the proof-carrying paper. The proof is the finite algebraic closure in Paper 13; this toolkit shows how to inspect the same closure digitally and by ordinar...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-14: Paper 14 - GR Boundary-Repair Curvature	77	This paper defines the CQECMLX substrate meaning of curvature as a boundary-repair demand. The closed result is a transport-ledger theorem: every route has a typed status, demonstrated rows carry zero repair score, n...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
SUMMARY-II-Folded-Strand-Physics: Summary Paper II — Folded Strand Physics: The Gluon as Quark, Mass, Curvature, Tower, and Moonshine	75	This paper presents the physics applications of the Gluon formalism. We do not use the word "Gluon" because the fit is good. We use it because the substrate IS SU(3), and the physics IS the **standard model ga...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	12 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series **Formalizing LCR, Unifying Standard Models**. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR - 32
Legacy source slot(s)	13
Ribbon role	order-2 slot-3: fold the initial action into a higher-dimensional representation
Proof form	fold map, coordinate atlas, and representation transport
Received state	state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport)
Produced state	formal result of original slot 13 plus the same-family lift path toward slot 23

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 32.1	standard_theorem_citation_bound_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD
Proposition 32.2	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	QCD color translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 32.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	physical quark confinement, CKM, and measured QCD parameters remain external calibration

Figure Plan

Figure	Purpose	Caption
FLCR-32-F1	Representation fold atlas	Schematic showing how FLCR - 32 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-32-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-32-F3	Same-family lift context	Diagram placing this paper in the original 00 - 79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-32-F1: Representation fold atlas

received state
-> Representation fold atlas
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

Table Plan

Table	Purpose
FLCR-32-T1	Face/representation correspondence table
FLCR-32-T2	Claim-lane/evidence-boundary table
FLCR-32-T3	Archive evidence card and source-backed upgrade table
FLCR-32-T4	Workbook replay and falsifier table

Worked Example

Map one source object across two labels/faces while preserving its address. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
atlas	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
face	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
fold	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.
representation transport	Defined by this paper as part of the <code>fold_action</code> proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-32-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-32 (QCD And Color Transport In LCR) occupies the `fold_action` role at lift depth 1. The paper receives state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport). It produces formal result of

original slot 13 plus the same-family lift path toward slot 23. The state transition is: fold the local state into a higher representation while preserving addressability.

The paper-local proof form is:

fold map, coordinate atlas, and representation transport

The integration result is a representation-transport chart with named invariants and residues. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
representation theory	Formal carrier for the paper-local result.
group actions and equivariance	Formal carrier for the paper-local result.
tensor or coordinate atlases	Formal carrier for the paper-local result.
transport maps between equivalent descriptions	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 32.1	standard_theorem_citation_bound_result	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD
Proposition 32.2	normal_form_result	QCD color translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 32.3	falsifier_or_open_obligation	physical quark confinement, CKM, and measured QCD parameters remain external calibration

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	13_Standard_Model_Quark_Face_Transport.md, supplements/SM_BRIDGE_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper’s results section is organized around five publication objects:

- 1. the exact object acted on at this slot;
- 2. the admissible operation or transformation;

3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-32-F1: Representation fold atlas	Visualizes the proof object, routing, or boundary.
FLCR-32-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-32-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.
Table	Role
FLCR-32-T1: Face/representation correspondence table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-32-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-32-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-32-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.